Aletheia: A Multi-Engine Retrieval-Augmented Architecture for Real-Time Hallucination Detection and Automated Fact Verification in Large Language Models

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Abstract - Large Language Models (LLMs) have achieved state-of-the-art capabilities across diverse Natural Language Processing (NLP) tasks. However, their tendency to generate factually incorrect, ungrounded, or plausible-sounding false information—commonly known as AI Hallucination—remains a major critical failure mode in high-stakes deployments. This paper introduces Aletheia an end-to-end framework designed to detect, quantify, and mitigate AI hallucinations in generated responses. Aletheia employs a 4-stage pipeline: (1) Atomic Claim Extraction, (2) Multi-Engine Evidence Retrieval (aggregating Wikipedia API, Web Search Engines, Academic/AI Repositories, and Local Vector Databases), (3) Semantic Entailment & Verdict Classification, and (4) Automated Revision with Auditable Ledger Logging. Our experimental evaluations indicate that multi-engine cross-verification improves hallucination detection recall by compared to single-source RAG architectures while maintaining sub-second inference overhead per atomic claim.

Keywords: Artificial Intelligence, Large Language Models, Hallucination Detection, Multi-Source Retrieval-Augmented Generation (RAG), Semantic Entailment, Natural Language Processing, Fact Verification, Knowledge Graphs.

1. Introduction

1.1 Background & Motivation

Large Language Models (LLMs) such as GPT-4, LLaMA-3, and Claude 3 rely on probabilistic autoregressive token generation. While effective for creative tasks, this architecture inherently risks producing output that lacks factual grounding in real-world knowledge. In domain-critical environments—such as medical diagnostics, legal analysis, and scientific research—hallucinated statements pose severe financial, ethical, and operational risks.

1.2 Taxonomy of AI Hallucinations

AI hallucinations are broadly classified into three categories:

1. **Intrinsic Hallucinations:** Direct contradictions with provided source material or ground truth.
2. **Extrinsic Hallucinations:** Assertions that can neither be verified nor refuted using the input context (unverifiable generation).
3. **Factual Drift:** Gradual departure from initial factual constraints during multi-turn conversational generations.

1.3 The Aletheia Solution

To systematically address these failure modes, we present **Aletheia (v1.0_FACTUALITY_CORE)**. Aletheia decouples token generation from truth verification by

applying a decoupled post-hoc and real-time auditing pipeline. By isolating AI responses into granular **atomic claims** and cross-referencing them against a federated network of search engines and reference corpora, Aletheia produces verifiable evidence, audit trails, and concrete text revisions.

2. Literature Review & Related Work

2.1 Retrieval-Augmented Generation (RAG)

RAG models ground generation by conditioning the language model on external documents retrieved via dense embedding search (e.g., FAISS, ChromaDB, Pinecone). However, standard RAG systems suffer from single-point-of-failure risks: if the single retriever fetches incomplete or biased data, the LLM incorporates that noise into its output.

2.2 Claim Decomposition & Granular Verification

Recent literature (e.g., Min et al., FActScore) demonstrates that evaluating long-form text as a whole yields low verification accuracy. Breaking down complex paragraphs into independent, verifiable **atomic propositions** significantly enhances precision in fact-checking models.

2.3 Limitations of Current Frameworks

Existing fact-checking frameworks primarily rely on a single knowledge source (usually Wikipedia or Google Web Search). Aletheia bridges this gap by establishing a federated multi-engine aggregation layer combining Wikipedia, Web Search, Academic Repositories (ArXiv, Semantic Scholar), and Local Private Databases.

3. System Architecture & Methodology

Aletheia operates on a modular, four-stage verification pipeline. The overall pipeline is depicted below:

INPUT MODULE

STAGE 1: ATOMIC CLAIM EXTRACTION

STAGE 2: MULTI-ENGINE EVIDENCE RETRIEVAL

Wikipedia API Web Search Engine Academic / AI Report

STAGE 3: SEMANTIC ENTAILMENT & VERDICT CLASSIFICATION Scoring: Entailment (P_E), Contradiction (P_C), Neutral (P_N)

STAGE 4: REVISION & AUDITABLE LEDGER EXPORT

Exporting Auditable Ledger, Corrected Output, and Source Attribution

3.1 Stage 1: Atomic Claim Extraction

Given a candidate response , Aletheia applies a rule-based syntactic parser coupled with an instruction-tuned claim extraction model . The objective is to decompose into a set of independent atomic claims:

Where each claim contains exactly one subject-predicate-object relation and carries no coreferences (pronouns are resolved to explicit entities).

3.2 Stage 2: Multi-Engine Evidence Retrieval

For each extracted claim , Aletheia executes concurrent, asynchronous API calls to four retrieval layers:

1. **Wikipedia API Engine ():** Fetches authoritative encyclopedic definitions, historical facts, and entity descriptions.
2. **Web Search Engine API ():** Utilizes Google/DuckDuckGo APIs for real-time, dynamic, and up-to-date facts.
3. **Academic / AI Corpus Engine ():** Queries ArXiv, Semantic Scholar, and OpenAlex APIs for scientific, mathematical, and machine learning assertions.
4. **Local Corpus Vector Store ():** Uses dense embedding similarity (Cosine Similarity on OpenAI text-embedding-3-small or bge-large-en) over user-provided reference metadata ().

The retrieved context passages for claim are formulated as:

3.3 Stage 3: Semantic Entailment & Verdict Classification

Each claim-passage pair is evaluated using a Natural Language Inference (NLI) Cross-Encoder model. The classifier outputs probability distributions over three canonical semantic states:

Where:

- **Entailment ():** The evidence directly supports the claim (Verdict: **VERIFIED**).
- **Contradiction ():** The evidence directly refutes the claim (Verdict: **HALLUCINATION**).
- **Neutral ():** The evidence neither supports nor refutes the claim (Verdict: **UNVERIFIED / INSUFFICIENT EVIDENCE**).

Mathematical Scoring Formula: The overall factuality confidence score for response is computed as:

Where represents the semantic weight of claim , and is the indicator function.

3.4 Stage 4: Automated Revision & Auditable Ledger

If a claim is flagged as a Contradiction (), Aletheia activates its Revision Module. The module synthesizes a corrected assertion using the retrieved context , substituting hallucinated entities with verified facts while retaining original stylistic integrity. All steps are logged into an **Auditable Evidence Ledger**.

4. Implementation Details & User Interface

Aletheia is implemented as a high-performance web platform:

- **Backend Stack:** FastAPI (Python 3.11), PyTorch, HuggingFace Transformers, LangChain, Asyncio for parallel retrieval.
- **Frontend Stack:** Next.js, Tailwind CSS, Lucide Icons, Dark-mode Factuality Interface.
- **Server Port Configuration:** Hosted on PORT:8000 with status monitor
- Verify with confidence.Aletheia decomposes AI-generated text into factual claims, retrieves verifiable evidence from your local corpus & search engines, and produces an auditable verification breakdown with suggested revisions.

01 / CREATE REVIEW HOW IT WORKS
Bring an answer into reviewEvidence, not guesswork

ORIGINAL QUESTION:

01 EXTRACT ATOMIC CLASS

02 RETRIEVE PASSAGES

03 CLASSIFY VERDICTS

ANSWER TO VERIFY:

04 REVISE & EXPORT

5. Experimental Setup & Results

5.1 Dataset & Benchmarks

We evaluated Aletheia using three open-source benchmark datasets for LLM factuality:

1. **TruthfulQA:** Testing common misconceptions and hallucinations.
2. **FEVER (Fact Extraction and VERification):** Benchmark for claim verification.
3. **HALU-Eval:** Domain-specific hallucination evaluation set (QA, Dialogue, Summarization).

5.2 Comparative Analysis

We compared Aletheia against three baselines: (1) Standard Vanilla LLM (GPT-3.5-Turbo), (2) Single-Source Wikipedia RAG, and (3) Single-Source Web RAG.

Model / System	Precision (%)
Vanilla GPT-3.5-Turbo	58.4
Single-Source (Wikipedia RAG)	76.4
Single-Source (Web Search RAG)	81.2
Aletheia (Multi-Engine Core)	**92.8**

5.3 Key Findings

- **Reduction in False Positives:** Combining Web, Wikipedia, and Academic sources reduced false hallucination flags by **34.5%** compared to Wikipedia-only verification.
- **Latency Efficiency:** Through asynchronous multi-threading (asyncio.gather), Aletheia achieves total execution latency under **1.8 seconds** for a standard 200-word candidate response.

6. Challenges & Limitations

1. **Temporal Information Drift:** Web search engines occasionally index unverified real-time news, introducing low-quality evidence into Stage 2.
2. **API Rate Limits & Costs:** Running parallel queries across multi-engine APIs increases network overhead and query costs.
3. **Paraphrasing Ambiguity:** Highly complex metaphors or implicit reasoning in LLM outputs sometimes confuse atomic claim extractors.

7. Future Work & Enhancements

- **Multi-Modal Hallucination Verification:** Extending Aletheia to evaluate vision-language outputs (e.g., GPT-4V, Gemini Pro Vision) against image and video evidence.
- **Knowledge Graph Integration:** Incorporating Neo4j Graph RAG for structured relational entity resolution.
- **Local LLM Fine-Tuning:** Training lightweight local NLI models (e.g., LLaMA-3-8B-Instruct) to run fully offline without external API dependencies.

8. Conclusion

In this paper, we presented **Aletheia (v1.0_FACTUALITY_CORE)**, an automated, multi-engine verification architecture designed to eliminate hallucination risks in AI-generated responses. By integrating atomic claim decomposition with a federated retrieval network (Wikipedia, Web Search, Academic Repositories, and Local Vector DBs), Aletheia creates a transparent, auditable, and

reliable verification framework. Empirical evaluations demonstrate that multi-engine cross-verification significantly outperforms traditional single-source grounding models.

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